Ashish Pathak, PhD

Assistant Professor,
Mechanical Engineering, IIT Jodhpur
Curriculum Vitae

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About me

I joined IIT Jodhpur as an Assistant Professor in Mechanical Engineering in May 2021.

I completed my B.Tech and M.Tech in Mechanical Engineering from IIT Kharagpur. During my undergraduate research with Prof. Ghosh Moulic, I became interested in numerical methods for modeling fluid flows, particularly those with discontinuities such as shocks and multiphase phenomena.

For my B.Tech thesis, I evaluated numerical schemes for shock modeling in an adaptive mesh framework. My M.Tech thesis involved developing an in-house code for simulating mixed convection film condensation, using coordinate transformation and a body-fitted mesh to address matrix stiffness, solved via perturbation methods.

I pursued my Ph.D. at the University of Massachusetts Dartmouth under Prof. Mehdi Raessi, focusing on two-phase flows interacting with moving rigid bodies. Due to significant interface deformation, I used the Volume-of-Fluid method on a fixed grid, avoiding remeshing and enabling large topological changes. This solver was applied to Ocean Wave Energy Converters, yielding key insights into dynamic similarity. As a Postdoctoral Fellow at UMass Dartmouth, I studied droplet dynamics from spray-wall interactions and developed a phase-change solver for evaporating, deforming interfaces.

Education

Year	School	Degree
2017	University of Massachusetts Dartmouth	PhD in Engineering and Applied Science
2011	IIT Kharagpur	M. Tech. in Thermal Science and Engineering
2011	IIT Kharagpur	B. Tech. in Mechanical Engineering

Research Interests

My research interests include many application areas of high-performance computing and multiphase flows, that include, but not limited to the following:

- Fluid-Structure Interaction
- Ocean Energy
- Surface tension driven flows in microfluidic devices
- Phase change problems
- High performance computing and visualization

Awards

Year	Award	Awarded by	Citation
2020	Milton Van Dyke Award	American Physical Society	Submission titled "Impact of high-speed diesel drop trains-pursuing cleaner diesel engines" at APS-DFD meeting, Chicago 2020
2016	Best Poster Award	CSCVR, UMass Dartmouth	Poster titled "Advanced computational simulations of ocean wave energy converters" at HPC Day, UMass Dartmouth 2016
2014	APS Travel Grant	American Physical Society	Attend APS-DFD meeting, San Francisco, 2014
2011	Institue Silver Medal	IIT Kharagpur	Securing highest CGPA at the end of 10th semester in

			Mechanical Engineering Dual Degree program
2011	Dwarka Nath Singh Memorial Prize	IIT Kharagpur	Best outgoing Dual Degree student in the Dept. of Mechanical Engineering scoring the highest CGPA at the end of 10th Semester among all the dual degree students of the department

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